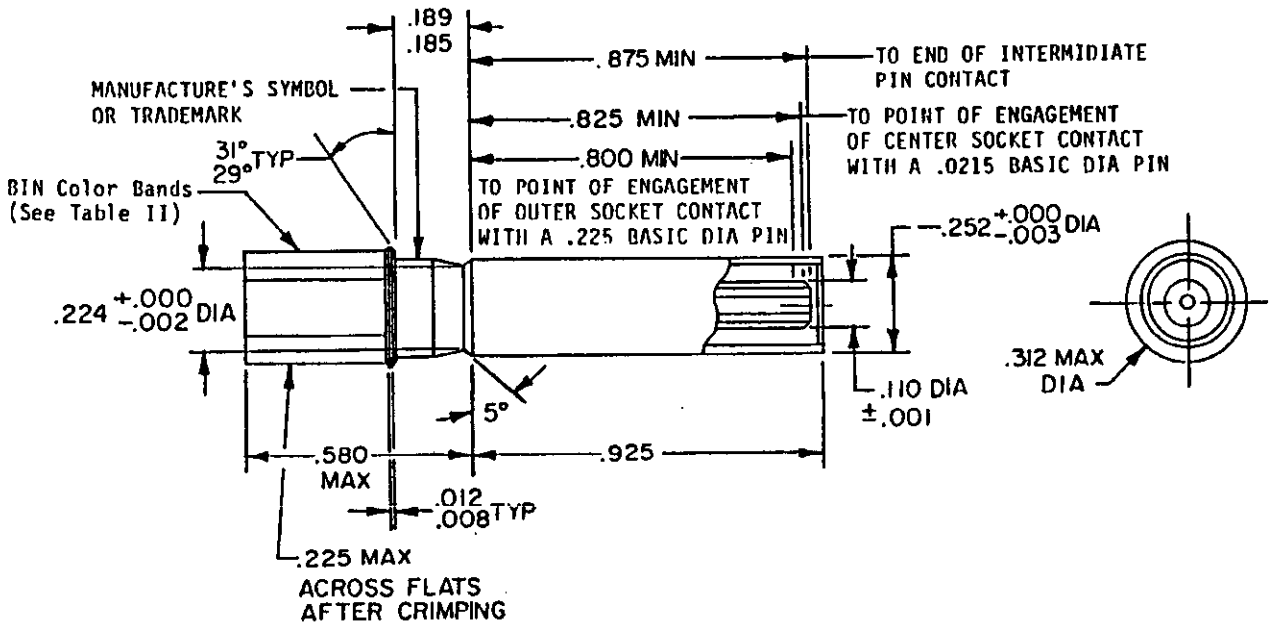
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, CONCENTRIC TWINAX, SOCKET, SHIELDED, SIZE 8 (FOR MIL-C-28840)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.



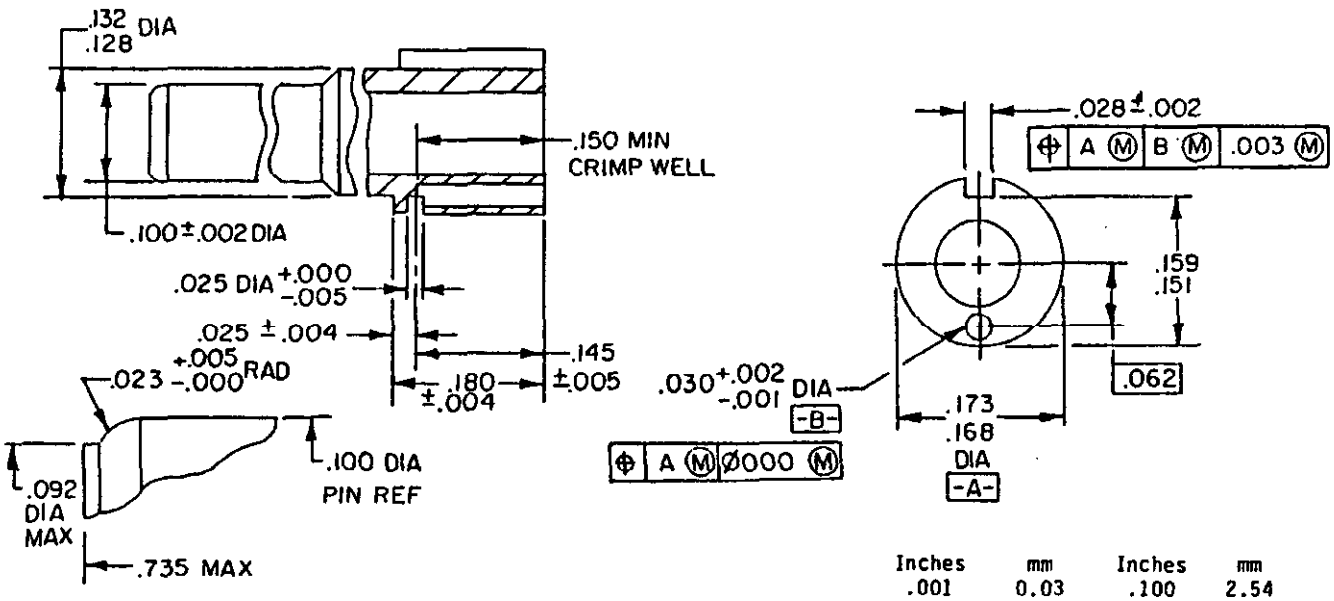
Inches	mm	Inches	mm	Inches	mm
.001	0.02	.110	2.79	.312	7.92
.002	0.05	.1115	2.83	.580	14.73
.0025	0.06	.185	4.69	.800	20.32
.003	0.07	.189	4.80	.825	20.95
.008	0.20	.224	5.68	.875	22.22
.012	0.30	.225	5.71	.925	23.49
.0215	0.54	.252	6.40		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Point at which a square ended pin of the same basic diameter as the mating contact first engages the intermediate contact spring. Provision for clearance hole shall be provided for the test pin.

FIGURE 1. Concentric twinax contact.

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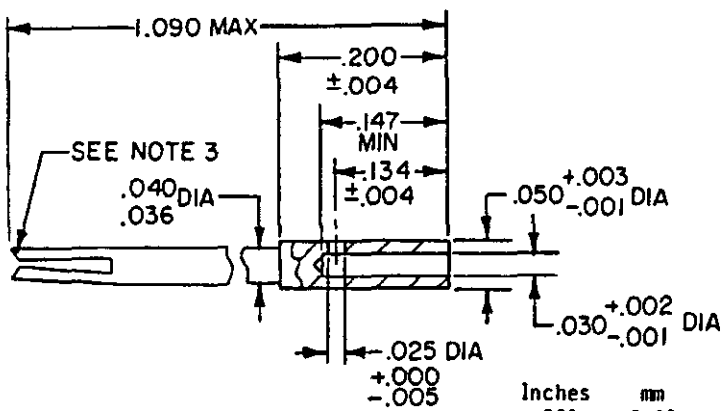
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.



Inches	mm	Inches	mm
.001	0.03	.100	2.54
.002	0.05	.128	3.25
.003	0.08	.132	3.35
.004	0.10	.145	3.68
.005	0.13	.150	3.81
.023	0.58	.151	3.84
.025	0.64	.159	4.04
.028	0.71	.168	4.27
.030	0.76	.173	4.39
.062	1.57	.180	4.57
.092	2.34	.735	18.67

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Intermediate pin contact.



Inches	mm	Inches	mm	Inches	mm
.001	0.03	.005	0.13	.050	1.27
.002	0.05	.025	0.64	.134	3.40
.003	0.08	.030	0.76	.147	3.73
.004	0.10	.036	0.91	.200	5.08
		.040	1.02	1.090	27.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Closed entry may be provided either by an integral part of contact design or by surrounding dielectric material.

FIGURE 3. Center socket contact.

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REQUIREMENTS:

Qualification:

Contacts shall comply with the reliability assurance provisions of MIL-STD-790 as specified in MIL-C-28840.

Design and construction:

Dimensions and configurations: See figures 1 through 3 and table I.

TABLE I. Design characteristics.

BIN code	Color bands			Contact cavity size	Cable accommodated	Type	Class
	1st	2nd	3rd				
557	Green	Green	Violet	8	M17/176-00002	D	B

Assembly procedure: Manufacturer's recommended assembly instructions shall be shipped with unit package.

Electrical:

Low signal level contact resistance (center and intermediate contacts only): See table II.

Contact resistance: See table III.

Frequency: 0 to 20 MHz (operating frequency range).

Voltage rating: 500 volts rms maximum; working voltage at sea level, 125 volts rms maximum at 70,000 feet.

TABLE II. Low signal level contact resistance (center and intermediate contacts only).

BIN code	Cable accommodated	Maximum contact resistance (milliohms)	
		Initial	After conditioning
557	M17/176-00002	55	66

TABLE III. Contact resistance.

BIN code	Contact	Cable accommodated	Test current (amperes)	Maximum voltage drop (millivolts)		
				+25°C, +3°C 0°C		+175°C, +3°C, 0°C
				Initial	After conditioning	After conditioning
557	Center	M17/176-00002	1.0	55	66	94
557	Intermediate	M17/176-00002	1.0	55	66	94
557	Outer	M17/176-00002	12.0	75	90	128

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Dielectric withstanding voltage: Shall be as specified in table IV.

TABLE IV. Dielectric withstanding voltage.

Contacts	Altitude	Test voltages ac rms
Center to intermediate	Sea level	1,000
Intermediate to outer	Sea level	500

Mating contact: Shall be in accordance with MIL-C-39029/104.

Mechanical:

Contact engagement and separation force (socket contacts only): The engagement depth shall be a minimum of 0.7 of the minimum socket bored depth. The test pins shall be in accordance with MS3197, except the diameters shall be as specified in table V. Provision for clearance hole on outer contact. Test pins shall be provided.

TABLE V. Contact engagement and separation force.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces)		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.2260 +.0000 -.0001	NA	NA	48	60	NA
.2240 +.0001 -.0000	3.0	2.0	NA	NA	NA
.0190 +.0000 -.0001	NA	NA	12	14	NA
.0210 +.0001 -.0000	0.5	0.4	NA	NA	NA

Crimp tensile strength (center, intermediate, and outer contact crimp joint): See table VI.

TABLE VI. Crimp tensile strength (at ambient).

BIN code	Cable accommodated	Axial load (pounds, minimum)		
		Center contact	Intermediate contact	Outer contact
557	M17/176-00002	8	8	25

Vibration: Method 2005 of MIL-STD-1344, test condition V. The following details shall apply:

a. Use the vibration envelope shown on figure 4.

b. Vibration: To be conducted at standard test conditions.

c. Duration: Eight hours in the longitudinal direction and 8 hours in a perpendicular direction (16 hours).

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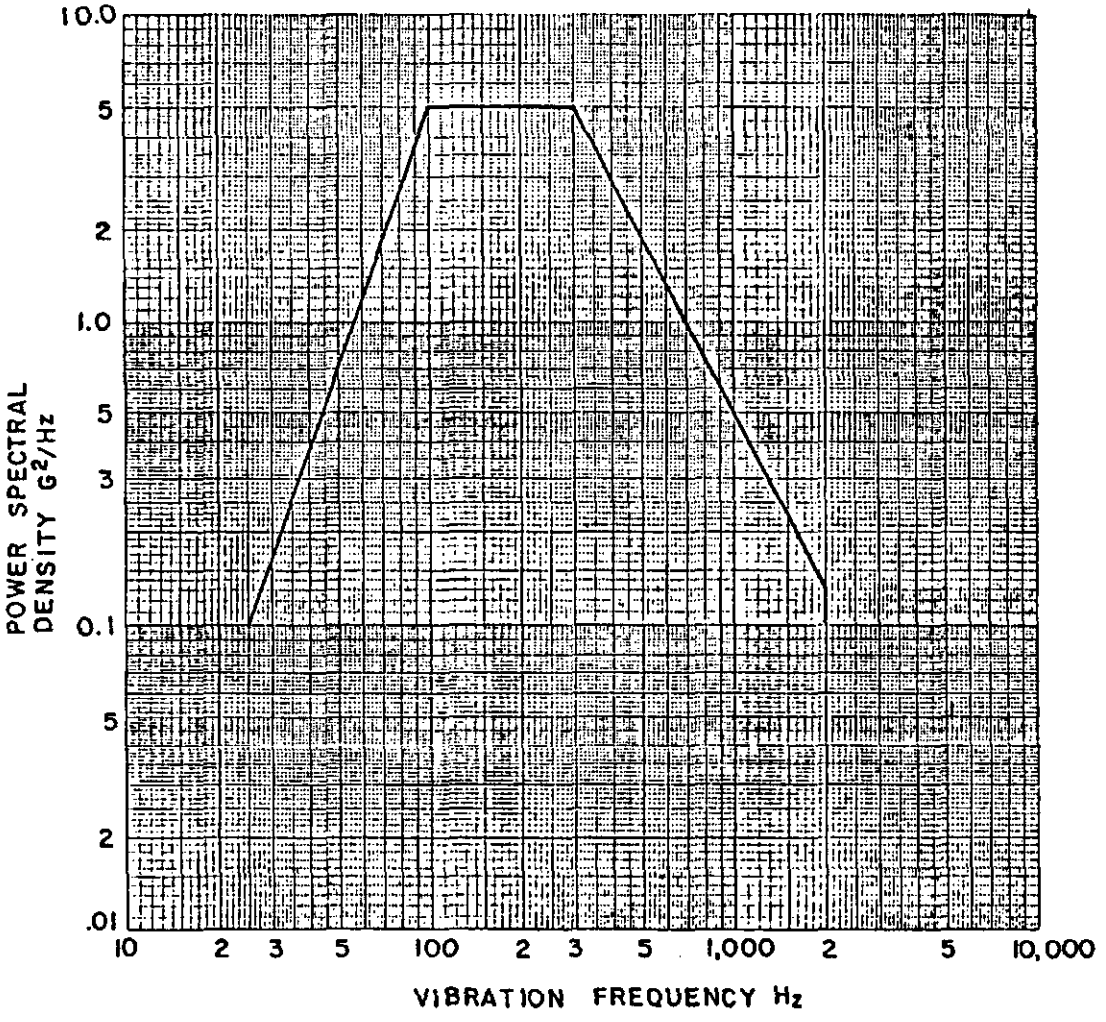


FIGURE 4. Vibration envelope.

High impact shock: All contacts shall be wired in a series circuit with 100 milliampers maximum current flow through the series circuit during shock. Contacts shall be monitored for any discontinuities. A detector capable of detecting all discontinuities in excess of 1 microsecond shall be used. Wired and mated contacts shall be subjected to the test specified in MIL-S-901, grade A with the following modifications and additions. Mounting fixture shall be in accordance with MIL-S-901. The cable or wire bundle shall be supported on a stationary frame to provide a free flexing cable length between the frame and fixture of not less than 36 inches (914.4 mm).

a. Coupling: Coupled by normal mounting means.

b. Test condition A: The wire bundle shall be provided with straight, open frame, strain relief accessory hardware.

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Tools: Shall be as specified in tables VII and VIII.

TABLE VII. Tools (installing and removal).

BIN code	Installing tool	Removal tool
557	M81969/17-06	M81969/19-03

TABLE VIII. Tools (crimping).

BIN code	Center contact tooling		Intermediate contact tooling		Outer crimp sleeve tooling	
	Basic crimping tool	Contact positioner 1/	Basic crimping tool	Die part number 1/	Basic crimping tool	Die part number 1/
557	M22520/2-01	K838	M22520/5-01	Y779	M22520/5-01	Y779

1/ Daniels Manufacturing Corporation part number is specified. Tools may be provided by other vendors if they are determined to be equivalent by the qualifying activity.

Environmental:

Temperature rating: -55°C to +200°C.

Humidity: In accordance with MIL-STD-1344, method 1002, type II.

Part number: See table IX.

TABLE IX. Part number.

BIN code	Part number
557	M39029/105-557

QPL evaluating activity: Naval Weapons Support Center, Crane, IN 47522.

Custodians:
Navy - EC
Army - CR
Air Force - 85

User activities:
Navy - OS
Air Force - 11

Review activities:
Navy - AS
Marine Corp - MC
DLA - ES

Preparing activity:
Navy - EC

Agent:
DLA - ES

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